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Dynamical Systems - MA3081

Solutions to Exercise Sheet 6

Exercise 6.1 (Structural stability)**.**

Consider the following dynamics on the torus $\mathbb{T}^2$ with coordinates $(x, y) \in [0,1] \times [0,1]$. For any of the following vector fields, discuss the structural stability of the induced dynamical systems on $\mathbb{T}^2$:

a) $f(x,y) = \begin{pmatrix} \sin\left(2\pi\left(x - \frac{1}{4}\right)\right)(1-\cos(2\pi y)) + \cos(2\pi y) + 1 \\ \sin(2\pi y) \end{pmatrix}$

b) $f(x,y) = \begin{pmatrix} \sin\left(2\pi\left(x - \frac{1}{4}\right)\right)(1-\cos(2\pi y)) + \cos(2\pi y) + 1 \\ \sin(2\pi x)\sin(2\pi y) \end{pmatrix}$.

Consider now the following dynamics on the sphere $\mathbb{S}^2$ with coordinates $\{(x,y,z)\,|x^2+y^2+z^2=1\}$. Discuss the structural stability of the induced dynamical system on $\mathbb{S}^2$ for the following vector field:

c) $f(x,y,z) = \begin{pmatrix} xz \\ yz \\ -(1-z^2) \end{pmatrix}$.

Solution.

a) We show that the given vector field is structurally stable. First, we find that $y' = 0$ is solved by $y = 0$ and $y = \frac{1}{2}$. Nonetheless, setting $y = \frac{1}{2}$, we obtain $x' = 2$. Conversely, in $y = \frac{1}{2}$, the first differential equation is $x' = -2\sin\left(2\pi\left(x - \frac{1}{4}\right)\right)$. Then, the only fixed point are

$$p_1 = \left(\frac{1}{4}, \frac{1}{2}\right) \quad \text{and} \quad p_2 = \left(\frac{3}{4}, \frac{1}{2}\right).$$

The Jacobian is of the form

$$J(x,y) = 2\pi \begin{pmatrix} \cos\left(2\pi\left(x-\frac{1}{4}\right)\right)(1-\cos(2\pi y)) & \left(\sin\left(2\pi\left(x-\frac{1}{4}\right)\right) - 1\right)\sin(2\pi y) \\ 0 & \cos(2\pi y) \end{pmatrix}$$

for any $(x,y) \in \mathbb{T}^2$. Then,

$$J(p_1) = 2\pi \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad J(p_2) = 2\pi \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix}.$$

This entails that p_1 is a saddle and p_2 is a stable fixed solution. They are also hyperbolic. We notice that the stable manifold of p_1 is $\left\{y = \frac{1}{2}\right\} \setminus \{p_2\}$, since $\left\{y = \frac{1}{2}\right\}$ is invariant. There are no homoclinic orbits in $[0,1] \times (0,1)$ due to the fact that y decreases for $1 > y > \frac{1}{2}$ and increases for $0 < y < \frac{1}{2}$. Furthermore, the set $\{y = 0\}$ is traveled by the (unique) periodic orbit. Its stability is already implied by the monotonicity of y in any neighbourhood, but it can also be studied by choosing the Poincaré section $\left\{(0, y) : \text{ for } 0 \leq y \leq \frac{1}{2}\right\}$. The projection along $\{x = 0\}$ of any orbit close to the axis $\{y = 0\}$ obeys to the second differential equation. In fact, linearizing on $(0, 0)$, we obtain the equation $\tilde{y}' = 2\pi\tilde{y}$. The orbit is then unstable and hyperbolic. In conclusion, Peixoto's Theorem implies structural stability.

Alternative: One can also determine stability of the periodic orbit by solving the variational equation. This leads to the fundamental solution

$$M(t) = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi t} \end{pmatrix}$$

and the Poincaré multiplier $e^{\pi} > 1$.

b) We now show that the vector field is structurally unstable. The equilibria of the system are

$$p_1 = \left(\frac{1}{4}, \frac{1}{2}\right) \quad , \quad p_2 = \left(\frac{3}{4}, \frac{1}{2}\right) \quad , \quad p_3 = \left(0, \frac{1}{4}\right) \quad \text{and} \quad p_4 = \left(0, \frac{3}{4}\right).$$

The Jacobian is

$$J(x, y) = 2\pi \begin{pmatrix} \cos\left(2\pi\left(x - \frac{1}{4}\right)\right)(1 - \cos(2\pi y)) & \left(\sin\left(2\pi\left(x - \frac{1}{4}\right)\right) - 1\right)\sin(2\pi y) \\ \cos(2\pi x)\sin(2\pi y) & \sin(2\pi x)\cos(2\pi y) \end{pmatrix}$$

for any $(x, y) \in \mathbb{T}^2$. We get then,

$$J(p_1) = 2\pi \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} \quad , \quad J(p_2) = 2\pi \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix},$$

$$J(p_3) = 2\pi \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad J(p_4) = 2\pi \begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}.$$

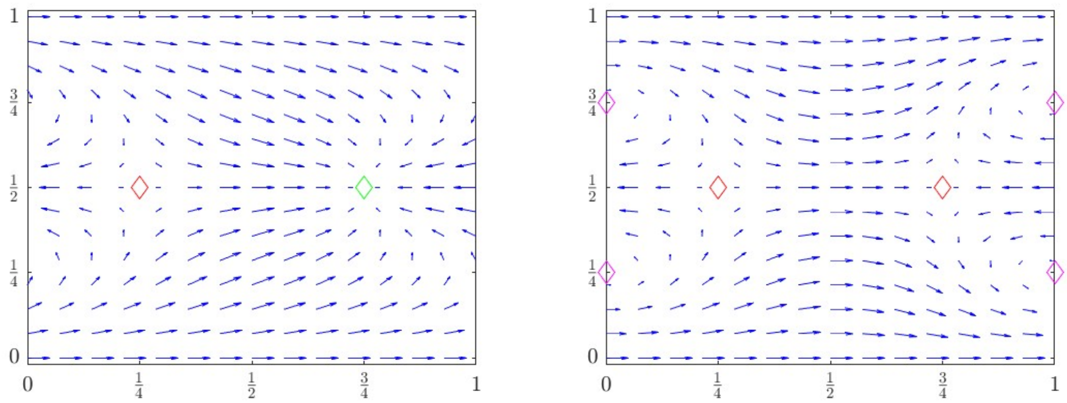
Peixoto's Theorem implies that the vector field is structurally unstable because p_3 and p_4 are centres.

Alternative: Notice that there are two heteroclinic orbits in $\{y = 0\}$ that start in p_1 and end in p_2. We can then apply Peixoto's Theorem since p_1 and p_2 are saddles.

Alternative: There are an infinite number of periodic orbits. This is implied by the fact that

$$f(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} f(1 - x, y)$$

for any $(x, y) \in \mathbb{T}^2$. It follows that any trajectory that crosses $\{x = 0\}$ and $\left\{x = \frac{1}{2}\right\}$ is periodic (it travels $\left\{x \in \left[0, \frac{1}{2}\right]\right\}$ and $\left\{x \in \left[\frac{1}{2}, 1\right]\right\}$ in the same time and the axis are crossed in the same points). By continuity of f and existence of a (hyperbolic) periodic orbit on $\{y = 0\}$, there exists in fact an infinite number of other periodic orbits in its neighbourhood. The conclusion follows, once again, from Peixoto's Theorem.



(a) Vector field for f in a)

(b) Vector field for f in b)

Figure 1: Green diamonds refer to sinks, red diamonds to saddles and magenta diamonds to centres. In (b), there are two centres, which are shown twice due to the periodic boundary.

c) The system can be written in spherical coordinates

$$\begin{aligned} x &= r\sin(\phi)\cos(\theta) \\ y &= r\sin(\phi)\sin(\theta) \\ z &= r\cos(\phi), \end{aligned}$$

for $0 \leq r$, $0 \leq \phi, \theta < 2\pi$. It becomes then

$$\begin{cases} \dot{r} &= \cos(\phi)(1-r^2) \\ \dot{\theta} &= 0 \\ \dot{\phi} &= \sin(\phi). \end{cases}$$

Due to the initial conditions and the invariance of $\mathbb{S}^2$, we set $r = 1$ and obtain

$$\begin{cases} \dot{r} &= 0 \\ \dot{\theta} &= 0 \\ \dot{\phi} &= \sin(\phi). \end{cases}$$

The Jacobian on $\mathbb{S}^2$ is

$$J(\theta, \phi) = \begin{pmatrix} 0 & 0 \\ 0 & \cos(\phi) \end{pmatrix}.$$

There are then only two steady states: $(1, 0, 0)$, a source, and $(1, 0, \pi)$, a sink. Respectively, they can be referred to in (x, y, z) coordinates as $(0, 0, 1)$ and $(0, 0, -1)$. The structural stability follows from Peixoto's Theorem.

Alternative: Of course, one can also find the equilibria $(0, 0, \pm 1)$ directly. Stability can be determined by either examining the sign of $\dot{z}$ or by examining $(\dot{x}, \dot{y})$ for $z \approx +1$ and $z \approx -1$.

$\square$

Exercise 6.2 (Hyperbolicity)**.** Prove that the set

$$H(\mathbb{R}^n) := \{A \in \mathbb{R}^{n\times n} \mid \mathrm{Re}(\lambda) \neq 0,\ \lambda \text{ eigenvalue for } A\}$$

is an open and dense subset of $\mathbb{R}^{n\times n}$.

Solution. In order to prove that $H(\mathbb{R}^n)$ is open we shall prove that for every element in $A \in H(\mathbb{R}^n)$ there is an open ball centered at A which is still contained in $H(\mathbb{R}^n)$. Denote by $\|\cdot\|$ any norm on $\mathbb{R}^{n\times n}$ (they all coincide since the space is finite-dimensional) and let $\lambda_1(A), \dots, \lambda_n(A)$ be the eigenvalues of A. Then, note that these eigenvalues of a matrix $A \in \mathbb{R}^{n\times n}$ are the roots of its characteristic polynomial, whose coefficients depend continuously on the entries of A. Since roots of a polynomial depend continuously on the coefficients of the polynomial, the eigenvalues of A depend continuously on A. Thus, there exists $\delta > 0$ such that for any $B \in \mathbb{R}^{n\times n}$ with $\|B - A\| < \delta$, it must be that $B \in H(\mathbb{R}^n)$. Consequently, $H(\mathbb{R}^n)$ is open.

In order to show that $H(\mathbb{R}^n)$ is dense, we shall firstly notice that given a matrix $A \in \mathbb{R}^{n\times n}$ and denoted by $\lambda_1, \dots, \lambda_n$ its eigenvalues, for every $\nu \in \mathbb{C}$ the matrix $A + \nu\,\mathrm{I}$ has eigenvalues $\lambda_1 + \nu, \dots, \lambda_n + \nu$. Now let $A \in R^{n\times n} \setminus H(\mathbb{R}^n)$. Then, there is $\varepsilon > 0$ such that the matrix $A_\varepsilon = A + \varepsilon\, I \in H(\mathbb{R}^n)$. Indeed, any eigenvalue of A with $\mathrm{Re}(\lambda) = 0$ will be "translated" into the eigenvalue $\lambda + \varepsilon$ of A_ε and $\mathrm{Re}(\lambda + \varepsilon) = \varepsilon \neq 0$, and if there are eigenvalues $\mu \in \mathrm{Eig}(A)$ such that $\mathrm{Re}(\mu) \neq 0$, then for ε sufficiently small, also $\mathrm{Re}(\mu + \varepsilon) = \mathrm{Re}(\mu) + \varepsilon \neq 0$. Finally, note that $\|A - A_\varepsilon\| = \|\varepsilon\, I\| \xrightarrow{\varepsilon \to 0} 0$, which concludes the proof.

□

Exercise 6.3 (Chain recurrence)**.**

Let X be a metric space and $g : X \to X$ a continuous map.

A periodic ε-pseudo-orbit is a finite ε-pseudo-orbit with $x_0 = x_n$ for some $n \in \mathbb{N}$. A point $x \in X$ is called chain recurrent if for every $\varepsilon > 0$ there is a periodic ε-pseudo-orbit from x to itself. Show that every non-wandering point is chain recurrent.

Solution. Let $x \in X$ be a non-wandering point and fix some arbitrary $\varepsilon > 0$. We have to show that there is an ε-pseudo-orbit from x back to x.

Since g is continuous there is some $\delta > 0$, such that $d(g(x), g(y)) < \varepsilon$ for all $y \in X$ with $d(x, y) < \delta$. Without restriction we can choose $\delta \leq \varepsilon$. Let $U := B_\delta(x)$ be the (open) ball of radius δ around x. Because x is non-wandering, there is some $n \in \mathbb{N}, n \geq 2$ such that $g^n(U) \cap U \neq \varnothing$, let us say $z \in g^n(U) \cap U$. Since $z \in g^n(U)$ there is some $y \in U$ with $z = g^n(y)$.

Now consider the sequence

$$\begin{aligned} x_0 &= x\,, \\ x_1 &= g(y)\,, \\ x_2 &= g^2(y) \\ &\dots \\ x_{n-1} &= g^{n-1}(y) \\ x_n &= x \end{aligned}$$

We have

$$d(g(x_0), x_1) = d(g(x), g(y)) \leq \varepsilon$$

by the choice of δ and since $y \in U = B_\delta(x)$. Furthermore,

$$d(g(x_{k-1}), x_k) = d(g^k(y), g^k(y)) = 0 \leq \varepsilon$$

for all $k = 1, 2, \ldots, n-2$ and finally

$$d(g(x_{n-1}), x_n) = d(g^n(y), x) = d(z, x) \leq \varepsilon$$

since $z \in U = B_\delta(x)$ and $\delta \leq \varepsilon$.

Consequently, the above sequence is indeed an ε-pseudo-orbit from x to x. As ε was arbitrary, we can conclude that x is chain recurrent. □

Exercise 6.4 (Perturbation of hyperbolic fixed points)**.**

Consider a C^1-map $f\colon \mathbb{R}^n \to \mathbb{R}^n$ with a hyperbolic fixed point $x_* \in \mathbb{R}^n$ and let $g \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ be arbitrary. Show that for sufficiently small ε, the perturbed map $f + \varepsilon g$ also has a hyperbolic fixed point with the same stability properties as x_*.

Hint: Use the implicit function theorem.

Solution. Define $F : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$, $F(x, \varepsilon) = f(x) + \varepsilon g(x) - x$. Then we have

$$F(x_*, 0) = f(x_*) + 0 \cdot g(x_*) - x_* = x_* - x_* = 0$$

because x_* is a fixed point of f. Furthermore,

$$D_x F(x, \varepsilon) = Df(x) + \varepsilon Dg(x) - \mathrm{Id}$$

so that

$$D_x F(x_*, 0) = Df(x_*) - \mathrm{Id}\,.$$

Since x_* is a hyperbolic fixed point, $Df(x_*)$ has no eigenvalue on the unit circle and consequently $Df(x_*) - \mathrm{Id}$ cannot have 0 as an eigenvalue, i.e. it is invertible. By the implicit function theorem there is a C^1-function $\psi : U \to \mathbb{R}^n$ defined in some neighbourhood $U \subset \mathbb{R}$ of 0 such that $\psi(0) = x_*$ and $F(\psi(\varepsilon), \varepsilon) = 0$ for all $\varepsilon \in U$. The equality $F(\psi(\varepsilon), \varepsilon) = 0$ is equivalent to saying that $\psi(\varepsilon)$ is a fixed point of $f + \varepsilon g$.

It remains to show hyperbolicity and (in)stability. To this end, note that $Df(\psi(\varepsilon)) + \varepsilon Dg(\psi(\varepsilon))$ depends continuously on ε and that the eigenvalues of a matrix depend continuously on the matrix. Thus, choosing ε sufficiently close to 0, we can assure that the multipliers of the perturbed system are still away from the unit circle, i.e. $\psi(\varepsilon)$ is also hyperbolic. To be more precise, the multipliers of the perturbed system will lie inside or outside unit circle like those of the unperturbed one and since stability for hyperbolic fixed points is determined only by the multipliers (cf. Hartman-Grobman Theorem) this property is maintained as well. □